Artificial Intelligence
Lecture 7: Agents based on propositional logic

Prof. Dr. Pierre-Alexandre Murena

SoSe

Outline

1 Reminders: Syntax and semantic of propositional logic

2 Agents based on propositional logic

3 Inference in propositional logic

Entailment

A sentence β is said to **entail** sentence α if and only if all models of β are also models of α:

$$M(\beta) \subseteq M(\alpha)$$

We then write:

$$\beta \models \alpha$$

Syntax: Backus-Naur Form of the grammar

$$
\begin{aligned}
< Proposition > \quad &\rightarrow \quad < AtomicSentences > \mid < CompoundSentences > \\
< AtomicSentences > \quad &\rightarrow \quad < PropositionalSymbols > \\
< CompoundSentences > \quad &\rightarrow \quad \neg(Proposition) \\
&\quad\quad\quad\quad \mid (< proposition >< binaryconnector >< proposition >) \\
< binaryconnector > \quad &\rightarrow \quad \wedge \mid \vee \mid \rightarrow \mid \leftrightarrow
\end{aligned}
$$

Semantics in model theory

Propositional symbols are atomic sentences of $\mathcal{PL}$: therefore a natural way is to attribute them a truth value, called an **interpretation**.

For all propositional variable x, we use the notations $\mathcal{I}(x)$, or $x^{\mathcal{I}}$ for an interpretation of x.

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An interpretation of a compound proposition is defined by giving an interpretation to connectors.

Truth value of connectors

$$(\neg\phi)^{\mathcal{I}} = \begin{cases} \mathit{false} & \text{if } \phi^{\mathcal{I}} = \mathit{true} \\ \mathit{true} & \text{otherwise} \end{cases}$$

$$(\phi_1 \wedge \phi_2)^{\mathcal{I}} = \begin{cases} \mathit{true} & \text{if } \phi_1^{\mathcal{I}} = \phi_2^{\mathcal{I}} = \mathit{true} \\ \mathit{false} & \text{otherwise} \end{cases}$$

$$(\phi_1 \vee \phi_2)^{\mathcal{I}} = \begin{cases} \mathit{false} & \text{if } \phi_1^{\mathcal{I}} = \phi_2^{\mathcal{I}} = \mathit{false} \\ \mathit{true} & \text{otherwise} \end{cases}$$

$$(\phi_1 \rightarrow \phi_2)^{\mathcal{I}} = \begin{cases} \mathit{true} & \text{if } \phi_1^{\mathcal{I}} = \mathit{false} \text{ or } \phi_2^{\mathcal{I}} = \mathit{true} \\ \mathit{false} & \text{otherwise} \end{cases}$$

$$(\phi_1 \leftrightarrow \phi_2)^{\mathcal{I}} = \begin{cases} \mathit{true} & \text{if } \phi_1^{\mathcal{I}} = \phi_2^{\mathcal{I}} \\ \mathit{false} & \text{otherwise} \end{cases}$$

Models and interpretations

A **model** of ϕ is an interpretation of ϕ which satisfies ϕ (i.e. $\mathcal{I}$ such that $\phi^{\mathcal{I}} = \mathit{true}$).

Example: The interpretation $\mathcal{I} = a\bar{b}$ is a model of $a \vee b$.
The models of $a \vee b$ are $\mathcal{M}(a \vee b) = \{ab, a\bar{b}, \bar{a}b\}$.

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Now that we have defined the models, we have also defined the entailment $\models$. This corresponds to defining the semantics of $\mathcal{PL}$. We use the notation $(\mathcal{PL}, \models)$ to designate the propositional logic endowed with this proposed semantic.

Computing the interpretation of a compound proposition

We can compute the interpretation of any proposition recursively:

Complexity 0: Propositional symbols are propositions of complexity 0. Their interpretation has been defined already.

Complexity $n+1$: A proposition of complexity $n+1$ is a proposition of the form $\neg(\phi_1)$ or $\phi_1 \cdot \phi_2$, where $\cdot$ is a binary connector and ϕ_1 and ϕ_2 are propositions of complexity n. The interpretation of a proposition of complexity $n+1$ is computed by combining the interpretation of its components,

Logical equivalence

Two propositions ϕ_1 and ϕ_2 are said to be **logically equivalent** if and only if they are true in the same models, i.e.:

$$\phi_1 \models \phi_2 \qquad \text{and} \qquad \phi_2 \models \phi_1$$

We write then $\phi_1 \equiv \phi_2$.

Valid propositions (tautologies)

A proposition is called **valid** if it is true for all models. We also call such a proposition a **tautology**.

Example: The proposition $\alpha \vee \neg\alpha$ is a tautology.

Deduction theorem

For any propositions α and β, $\alpha \models \beta$ if and only if $\alpha \rightarrow \beta$ is a tautology.

Substitution in formulas

The substitution of a propositional symbol x by a propositional formula ψ in a propositional formula ϕ is the propositional formula, denoted by $\phi[x \setminus \psi]$, similar to ϕ, but with all instances of x replaced by ψ.

Substitution Theorem

If ϕ is a tautology, then any substitution $\phi[x \setminus \psi]$ (for any propositional symbol x and any proposition ψ) is a tautology.

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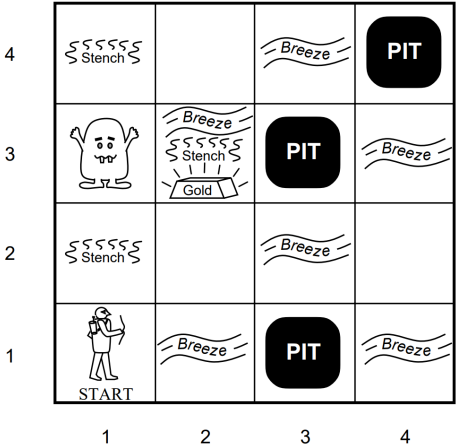
Back to the Wumpus world



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Representing the environment in the Wumpus world

We can represent the content of the cell (x, y) using propositional variables:

- $W_{x,y}$: indicates that the cell contains a Wumpus
- $G_{x,y}$: indicates that the cell contains gold
- $P_{x,y}$: indicates that the cell contains a pit
- $B_{x,y}$: indicates that the cell has breeze
- $S_{x,y}$: indicates that the cell has stench

The environment is fully described by $5n^2$ variables.

Remark: These variables don't evolve during the game. We call them **atemporal variables**.

Possible environments

A given configuration of the environment is associated to a tuple $(\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S})$.

Conversely, given $(\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S})$, can we reconstruct a valid environment?

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Conversely, given $(\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S})$, can we reconstruct a valid environment?

No! The environment is constrained by the rules of the game:

- Existence of one single Wumpus;
- A cell can contain at most the Wumpus, a pit or gold;
- Breeze only around a pit;
- Stench only around the Wunpus.

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We have to represent these rules in the language of propositional logic.

Existence of the Wumpus

Rule 1: There is a Wumpus.

How to represent Rule 1 in propositional logic using $(\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S})$?

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How to represent Rule 1 in propositional logic using $(\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S})$?

We can reformulate the rule in English: There is a Wumpus either in (1,1), or in (1,2), or in (1,3) etc.

Solution:

$$R_1 = W_{1,1} \vee W_{1,2} \vee \ldots \vee W_{n,n-1} \vee W_{n,n}$$

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Reformulation of Rule 2: For any $(x, y) \neq (x', y')$, there cannot be a Wumpus both in cell (x, y) and in cell (x', y').

Problem: There is no direct way of writing "for any" in propositional logic!

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Problem: There is no direct way of writing "for any" in propositional logic!

Solution 1: Split this rule into several rules:

$$R_2^{(1,1,1,2)} = W_{1,1} \rightarrow \neg W_{1,2}$$
$$R_2^{(1,1,1,3)} = W_{1,1} \rightarrow \neg W_{1,3}$$
$$\dots$$

Solution 2: Conjuction of rules:

$$R_2 = (W_{1,1} \rightarrow \neg W_{1,2}) \wedge (W_{1,1} \rightarrow \neg W_{1,3}) \wedge \dots \wedge (W_{n,n} \rightarrow \neg W_{n,n-1})$$

Remark: Knowledge bases in propositional logic

A **knowledge base** was introduced as a set of sentences (here propositional sentences):

$$KB = \{\phi_1, \ldots, \phi_K\}$$

The **models** of KB are the interpretations that make all ϕ_k true.

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The **models** of KB are the interpretations that make all ϕ_k true.

To describe the unicity of the Wumpus, we have proposed two solutions:

$$KB_1 = \left\{ R_2^{(1,1,1,2)}, \ldots, R_2^{(n,n,n-1)} \right\}$$
$$KB_2 = \left\{ R_2^{(1,1,1,2)} \wedge \ldots \wedge R_2^{(n,n,n-1)} \right\}$$

We can see that the models of KB_1 and of KB_2 are the same. In other words, in propositional logic, we can assimilate a knowledge base $\{\phi_1, \ldots, \phi_K\}$ with the conjunction $\phi_1 \wedge \ldots \wedge \phi_K$ of its sentences.

Unicity of the content of a cell

Rule 3: A cell can contain at most the Wumpus, a pit or gold

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Solution: Split this rule into several rules:

$$
\begin{aligned}
R_3^{(1,1,w)} &= W_{1,1} \rightarrow \neg G_{1,1} \wedge \neg P_{1,1} \\
R_3^{(1,1,g)} &= G_{1,1} \rightarrow \neg W_{1,1} \wedge \neg P_{1,1} \\
R_3^{(1,1,p)} &= P_{1,1} \rightarrow \neg W_{1,1} \wedge \neg G_{1,1} \\
R_3^{(1,2,w)} &= W_{1,2} \rightarrow \neg G_{1,2} \wedge \neg P_{1,2} \\
&\dots
\end{aligned}
$$

Breeze only around a pit

Rule 4: There is breeze only in the cells adjacent to a pit.

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Solution: For all (x, y), and ignoring the border effects:

$$R_4^{(x,y)} = B_{x,y} \leftrightarrow (P_{x-1,y} \vee P_{x+1,y} \vee P_{x,y-1} \vee P_{x,y+1})$$

At the borders, the rule can be adapted. For instance:

$$R_4^{(1,1)} = B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$$

Stench only around the Wumpus

Rule 5: There is breeze only in the cells adjacent to a pit.

Stench only around the Wumpus

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Solution: For all (x, y), and ignoring the border effects:

$$R_5^{(x,y)} = S_{x,y} \leftrightarrow (W_{x-1,y} \vee W_{x+1,y} \vee W_{x,y-1} \vee W_{x,y+1})$$

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Intuition: Representing the perceptions

Intuition: We can represent the perceptions of the agent by introducing variables *Stench* and *Breeze*, indicating whether the agent feels a stench or a breeze.

Problem: In practice, this does not work:

If the agent is in a cell and feels a breeze, it will add *Breeze* to its knowledge base.

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The knowledge base now contains *Breeze* and $\neg$*Breeze*, which makes it inconsistent!

The problem is that the percept asserts something only about the current time. Unlike the environment, they change over time. We call them **fluent**.

Temporal representation of percepts

To solve this problem, we associate the fluent properties with the time step. For instance:

$$Breeze^t$$

Now, a knowledge base containing $Breeze^t$ and $\neg Breeze^{t+1}$ is consistent.

The full perception of the agent at time t is then the conjunction of 5 variables:

$$(\neg)Gold_t \wedge (\neg)Wumpus_t \wedge (\neg)Pit_t \wedge (\neg)Breeze_t \wedge (\neg)Stench_t$$

Location of the agent

Unlike the Wumpus who stays still, the agent moves over time. Its position has to depend on time too:

Propositional variable $A^t_{x,y}$ indicates that the agent is in cell (x, y) at time t.

Connection between the environment and the perceptions

$Breeze^t$ alone only indicates that the agent perceives breeze at time t. This is a consequence of the location of the agent and of the configuration of the environment.

We add the following rules to connect the configuration of the environment and the perceptions of the agent, for all (x, y) and t:

$$A^t_{x,y} \rightarrow (Breeze^t \leftrightarrow B_{x,y})$$
$$A^t_{x,y} \rightarrow (Stench^t \leftrightarrow S_{x,y})$$
$$A^t_{x,y} \rightarrow (Pit^t \leftrightarrow P_{x,y})$$
$$A^t_{x,y} \rightarrow (Gold^t \leftrightarrow G_{x,y})$$
$$A^t_{x,y} \rightarrow (Wumpus^t \leftrightarrow W_{x,y})$$

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Actions

The actions are represented as propositional variables indexed by time:

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The **effect axioms** specify the outcome of an action at the next time step. For instance:

$$A^t_{x,y} \wedge Right^t \rightarrow A^{t+1}_{x,y+1}$$

The frame problem

Suppose now that the agent has the possibility to kill the Wumpus. Therefore, we add a fluent variable: $WumpusAlive^t$.

Obviously, by just moving, the agent does not modify the value of *WumpusAlive*. However, if this is not specified, there is no way the agent can infer this!
This is called the **frame problem**: the effect axioms fail to state what remains unchanged.

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Remark: When the number of actions and of fluents is high, this leads to a very large number of frame axioms...

Successor-state axioms

Solution: Focus on fluents rather than on actions:

- Either an action made the fluent get its value;
- Or the fluent had already this value, and the action did not change it.

Example: The agent is in cell $(1,1)$ at $t+1$ if:

- It was in an adjacent cell and did the right move;
- It was already in $(1,1)$ and bumped on a wall.

$$A_{1,1}^{t+1} \leftrightarrow (A_{2,1}^{t} \wedge Left^{t}) \vee (A_{1,2}^{t} \wedge Down^{t}) \vee (A_{1,1}^{t} \wedge (Down^{t} \vee Left^{t}))$$

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The search problem

Previously, we described the planning problem using the following formalism:

- **States:** What does the environment look like
- **Initial state:** How is the environment at the beginning
- **Goal:** Which states does the agent target
- **Actions:** Possible actions
- **Transition model:** Effect of the actions on the state

A propositional formulation of the search problem

We can use the formalism of propositional logic to specify the search problem

- **States:** Propositional representation of the environment and agent (atemporal and fluent variables)
- **Initial state:** Assertions about the initial state
- **Goal:** Assertions about the goal at time T
- **Actions:** Propositional variables for actions at each time up to T
- **Transition model:** Successor-state axioms at each time up to T

A sentence for the problem

Example:

- **States:** Rules $R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5 \wedge R_6$ and description $\mathbf{W}, \mathbf{G}, \mathbf{P}, \mathbf{B}, \mathbf{S}$
- **Initial state:** $A^0_{1,1}$
- **Goal:** $\bigwedge_{x,y} A^T_{x,y} \leftrightarrow G_{x,y}$
- **Actions:**
- **Transition model:** Successor-state axioms for the F fluent variables: $S_1 \wedge \ldots \wedge S_F$

$$P = \mathbf{W} \wedge \mathbf{G} \wedge \mathbf{P} \wedge \mathbf{B} \wedge \mathbf{S} \wedge R_1 \wedge \ldots \wedge R_6 \wedge A^0_{1,1} \wedge \left(\bigwedge_{x,y} A^T_{x,y} \leftrightarrow G_{x,y} \right) \wedge S_1 \wedge \ldots \wedge S_F$$

How to solve the planning problem?

We have reduced the description of the planning problem to a sentence in propositional logic. Can we use propositional logic to get a solution?

Idea: If a solution is the sequence of actions $Right, Up, Up, \ldots Left, Down$, then $P^{\mathcal{I}} = True$ for any interpretation $\mathcal{I}$ of P such that:

$$\begin{aligned}
&\mathcal{I}(Right_0) = True \\
&\mathcal{I}(Up_1) = True \\
&\mathcal{I}(Up_2) = True \\
&\quad \ldots \\
&\mathcal{I}(Left_{T-1}) = True \\
&\mathcal{I}(Down_T) = True
\end{aligned}$$

Planning as a satisfiability problem

The planning here corresponds to finding an interpretation (= a model) of the formula P or, equivalently, to checking that P is satisfiable.

This kind of problems is known as **SAT problem**:
Given a propositional formula ϕ, SAT consists in finding a model that satisfies ϕ.

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A simple way to solve a SAT problem is to test all interpretations. When ϕ contains n propositional variables, this means testing at most 2^n interpretations. Can we do better?

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Cook-Levin theorem

The Boolean satisfiability problem (SAT) is NP-complete.

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Planning in the Wumpus world

The problem of planning in the Wumpus world is that the agent does not know the position of the gold, of the Wumpus and of the pits. It has to **infer** them.

Solving the Wumpus game requires to:

(1) **Estimate** the state of the environment by reasoning about acquired knowledge;

(2) **Plan** where to go next to explore efficiently.

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(1) **Estimate** the state of the environment by reasoning about acquired knowledge;

(2) **Plan** where to go next to explore efficiently.

Task (1) is achieved with the tools offered by logic, while Task (2) is optimally solved by problem solving and planning.

A simple hybrid agent

The simplest version of an agent in the Wumpus world would work as follows:

(1) Estimate which cells are safe to enter

(2) Select an unvisited safe cell (if any)

(3) Plan route to the cell

(4) Modify route to get the gold if any evidence of its presence.

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Question: How to infer that a cell is safe?

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Inference

We have seen that the semantic of propositional logic allows to define the notion of entailment: $KB \models \alpha$.

However, just because $KB \models \alpha$ does not mean that we know that result!

The process of finding α such that $KB \models \alpha$ is called **inference**. Inference is a mechanism used to derive sentences from other sentences.

We use the notation

$$KB \vdash_i \alpha$$

to say that the sentence α is **derived** from KB by inference procedure i.

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- Propositional inference: Enumeration method
- Inference and proofs

Principle

$KB \models \alpha$ means that all models of KB are models of α.

Consequently, to check whether $KB \models \alpha$, it is enough to compute the truth of KB and of α for all interpretations, and to check that the interpretations that make KB true also make α true.

Example: Given the knowledge base $KB = (A \vee C) \wedge (B \vee \neg C)$ and $\alpha = A \vee B$. Show that $KB \models \alpha$.

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   - Propositional inference: Enumeration method
   - Inference and proofs

What is a proof?

A **proof** is a chain of conclusions that leads from the whole KB to the proposition α to infer.

These conclusions are obtained by applying some specific rules, called **inference rules**. We will now detail a few important inference rules.

Modus ponens

$$\frac{\begin{array}{c}\alpha \\ \alpha \rightarrow \beta\end{array}}{\beta}$$

For instance, if we have $(S_{1,1} \wedge \neg W_{1,2}) \rightarrow W_{2,1}$ and $(S_{1,1} \wedge \neg W_{1,2})$, then we can infer $W_{2,1}$.

And-elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

For instance, if we have $W_{2,1} \wedge \neg W_{1,2}$, then we can infer $W_{2,1}$.

Logical equivalences

We can derive inference rules from any logical equivalence.

For instance, knowing that $\alpha \leftrightarrow \beta \equiv (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$, we can derive the following inference rule:

$$\frac{\alpha \leftrightarrow \beta}{(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)}$$

Proof by resolution

$$\frac{\begin{array}{c} a_1 \vee \ldots \vee a_m \vee c \\ b_1 \vee \ldots \vee b_n \vee \neg c \end{array}}{a_1 \vee \ldots \vee a_m \vee b_1 \vee \ldots \vee b_n}$$

Remarks:

- Modus ponens can be seen as a special case of resolution
- Even though it may sound very specific, the resolution rule is very general: indeed, it can be shown that any propositional formula can be written in **conjunctive normal form**, i.e. as a conjunction of disjunctions of variables or of their negation.

Proof as planning

Surprisingly, we can describe proofs as a search problem:

- **States:** Knowledge bases
- **Initial state:** The initial knowledge base
- **Goal:** A knowledge base containing the sentence to prove
- **Actions:** All the inference rules applied to all the sentences that match the premises of the rules
- **Transition model:** Adding the sentence obtained by the inference using the chosen rule